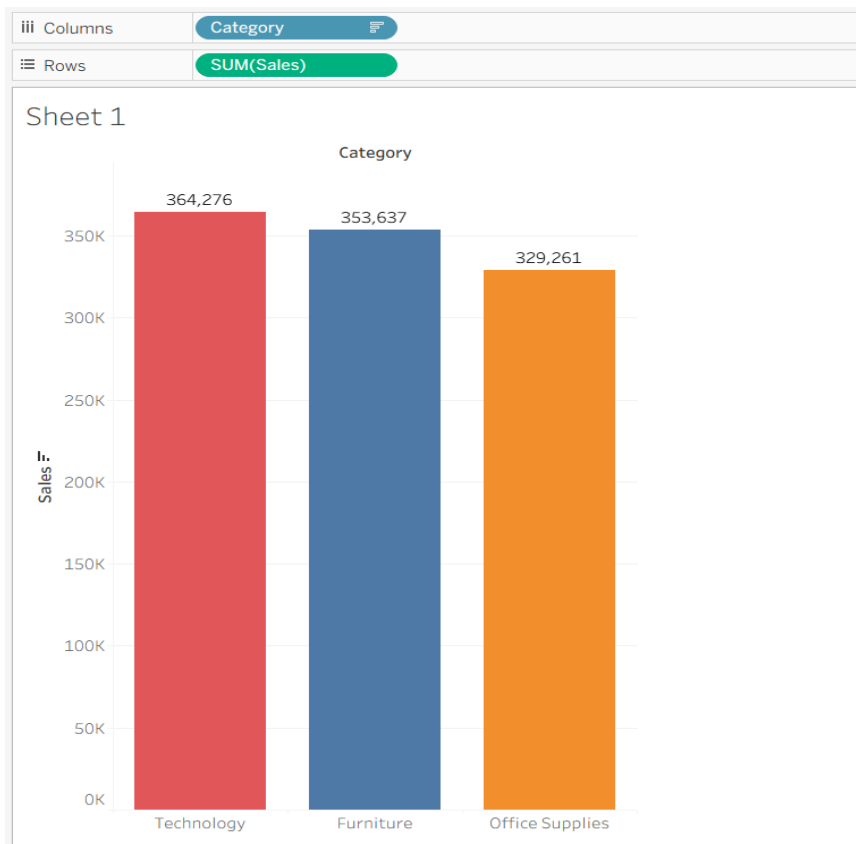


Tableau Visualizations Assignment

Task 1: Category Performance Analysis Business Need: Management wants to know which product categories generate the highest revenue. Task: Create a visualization to compare total Sales by Category and identify the top-performing category.

Answer:

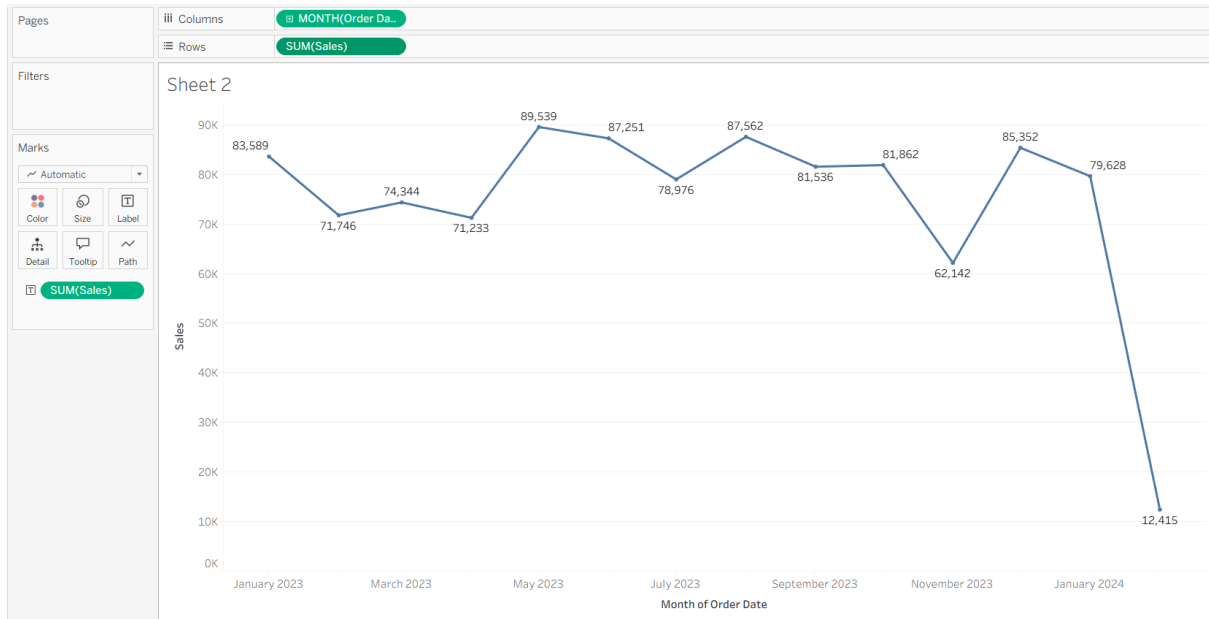
Technology is the top performing category generating \$364,276 in total sales — contributing 34.8% of total revenue. All three categories are relatively close in revenue — within a ~\$35,000 range of each other.



Task 2: Sales Trend Analysis Business Need: The leadership team wants to understand how sales fluctuate over time to plan promotions and inventory. Task: Analyze Sales trends over time (monthly) to identify peak and slow periods.

Answer:

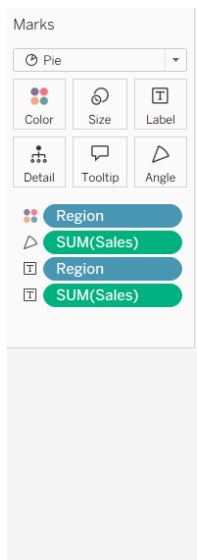
Your data clearly shows **May to August as the peak sales window** and **November as the weakest month** leadership should plan inventory and promotions around these patterns to maximize revenue throughout the year.



Task 3: Regional Contribution Analysis Business Need: The regional managers want clarity on how much each region contributes to overall sales. Task: Visualize Region-wise Sales contribution and identify the highest contributing region.

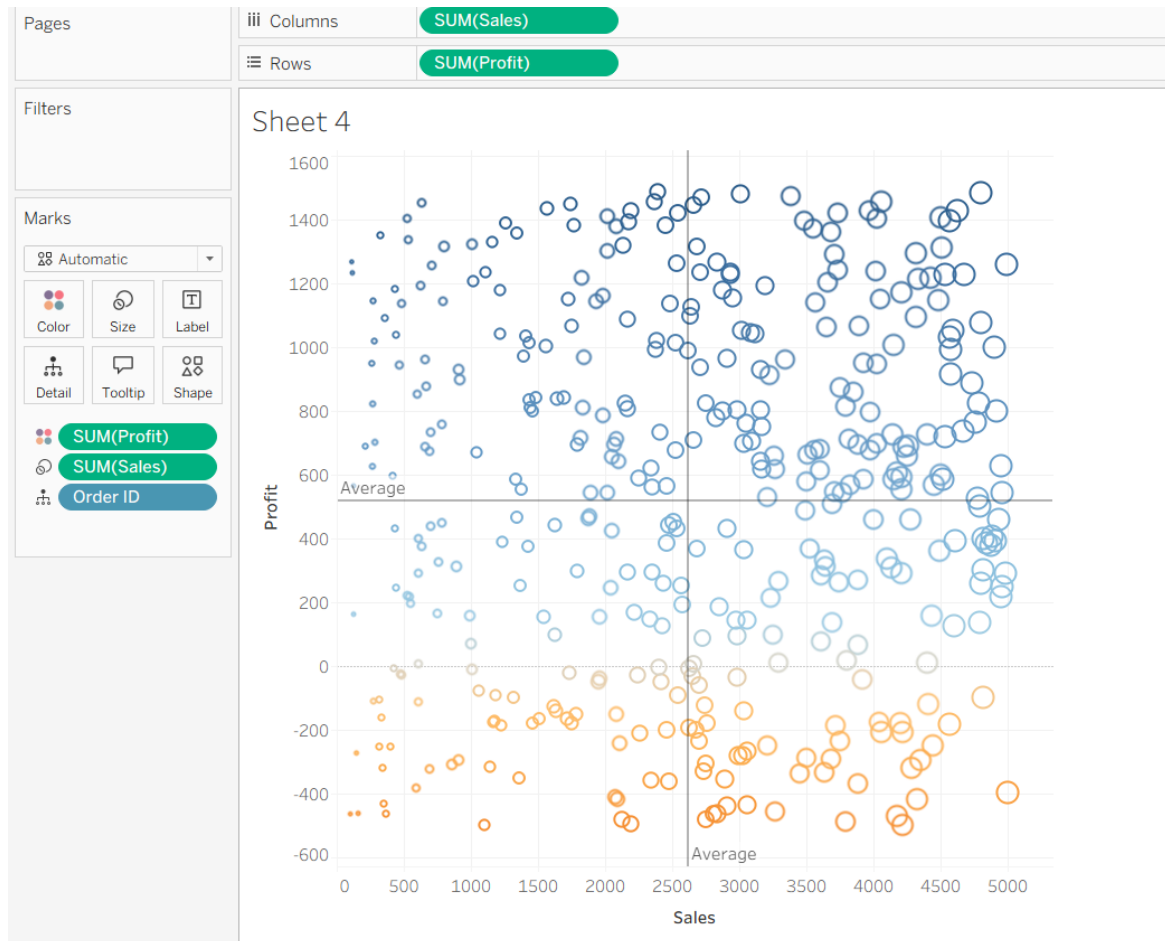
Answer:

West region is the clear leader in both Sales (\$287K) and Profit (\$63K) but all four regions are remarkably balanced within a 4.4% range of each other, suggesting a healthy and evenly distributed business across all regions.



Task 4: Profitability Investigation Business Need: The finance team suspects that some high-sales orders may be reducing overall profit. Task: Analyze the relationship between Sales and Profit to identify loss-making or low-profit orders.

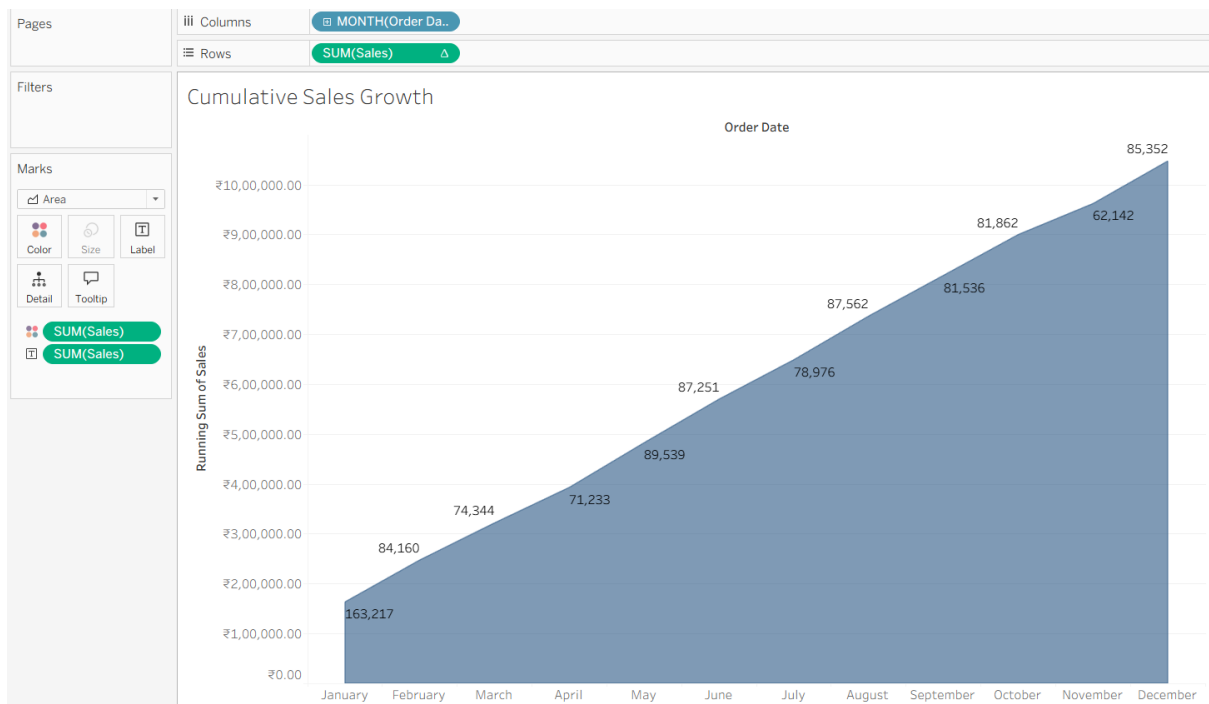
Answer:



Task 5: Business Growth Evaluation Business Need: Senior management wants to assess whether the business is growing consistently. Task: Visualize cumulative Sales growth over time to evaluate overall performance.

Answer:

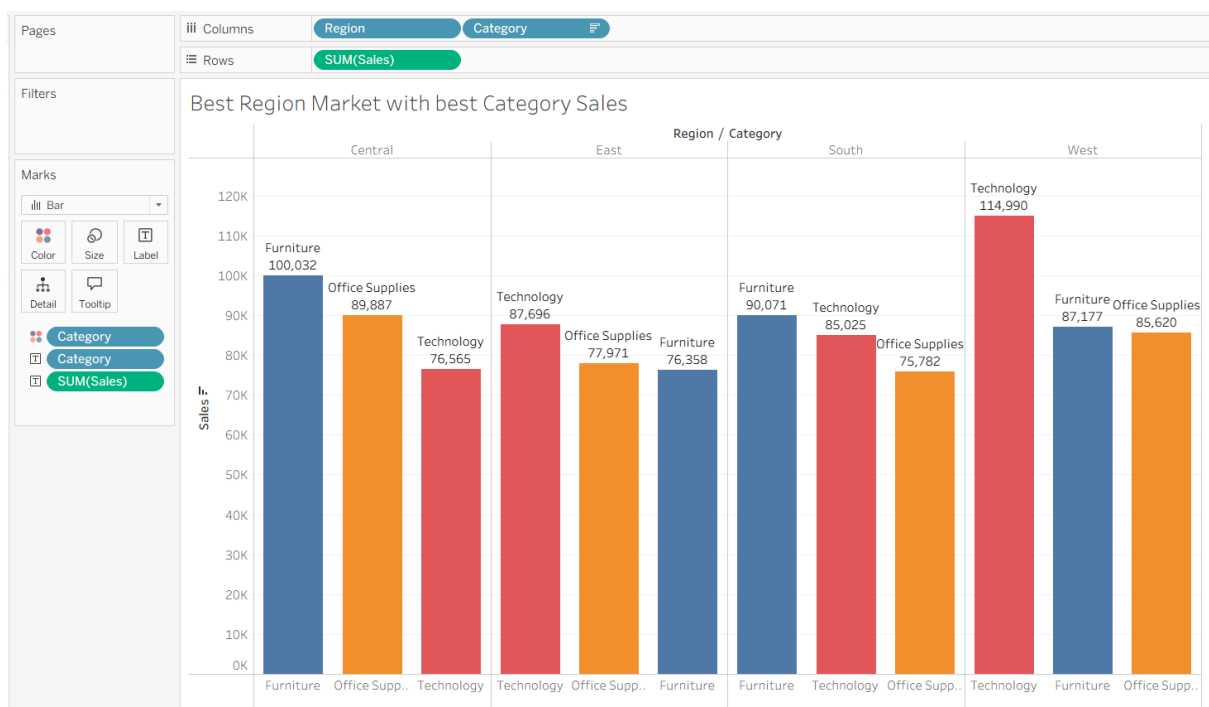
The business shows consistent cumulative growth crossing \$1 Million in just 13 months with May, August and December being the strongest months and November being the only significant weak spot that needs a targeted promotional strategy.



Task 6: Category vs Region Performance Business Need: Operations wants to know which regions perform well for each product category. Task: Compare Category-wise Sales across Regions to identify strong and weak market areas.

Answer:

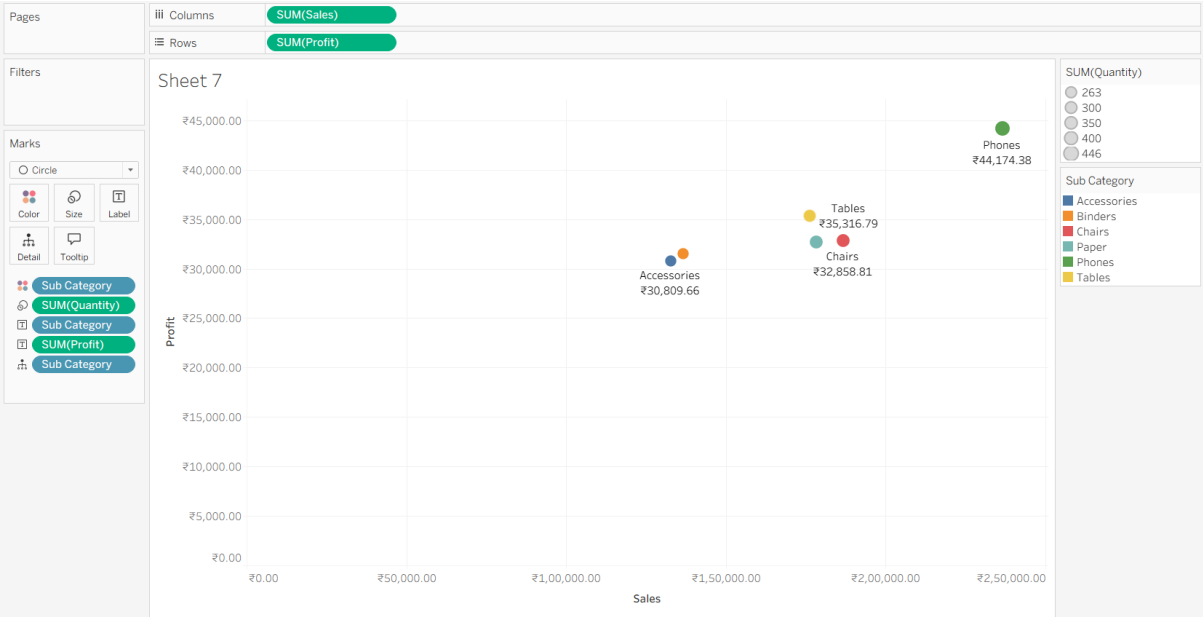
West + Technology is the undisputed champion at \$114,990 in sales while **South + Office Supplies** is the weakest link at \$75,782, giving Operations a clear picture of **where to push harder and where to rethink strategy**.



Task 7: Product Prioritization Business Need: The product team wants to focus on products that create maximum business impact. Task: Perform a multi-dimensional analysis using Sales, Profit, and Quantity to identify priority products.

Answer:

Phones is the undisputed priority product — leading in Sales, Profit and Quantity simultaneously — while **Accessories and Binders** are hidden gems with the best profit margins that deserve more marketing attention to unlock their full potential.

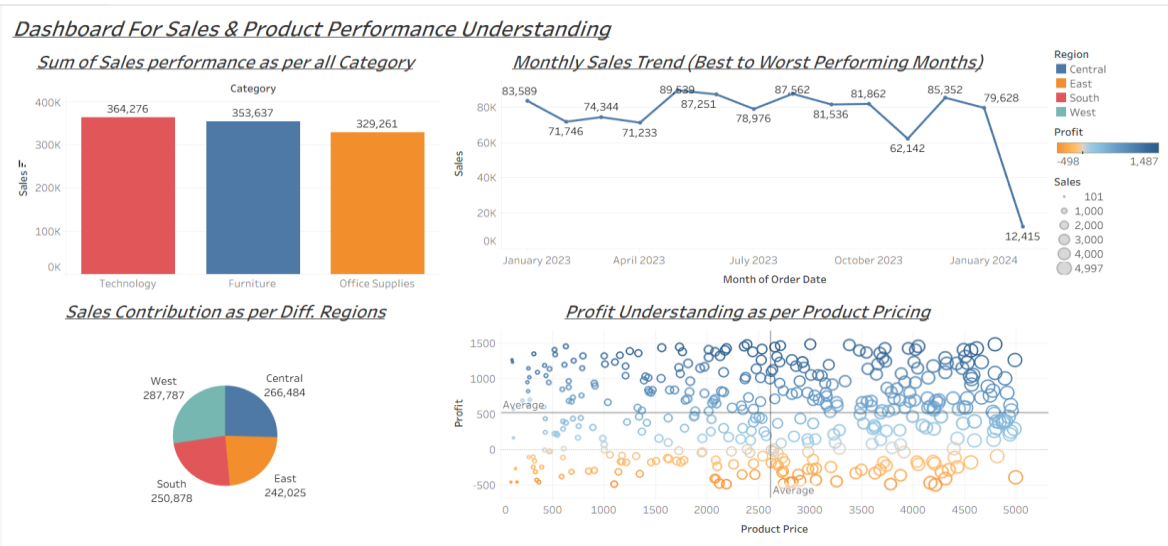


Task 8: Executive Dashboard Creation Business Need: The CEO wants a single view dashboard that summarizes key insights. Task: Create an interactive dashboard combining at least three visualizations and apply filters such as Region or Category.

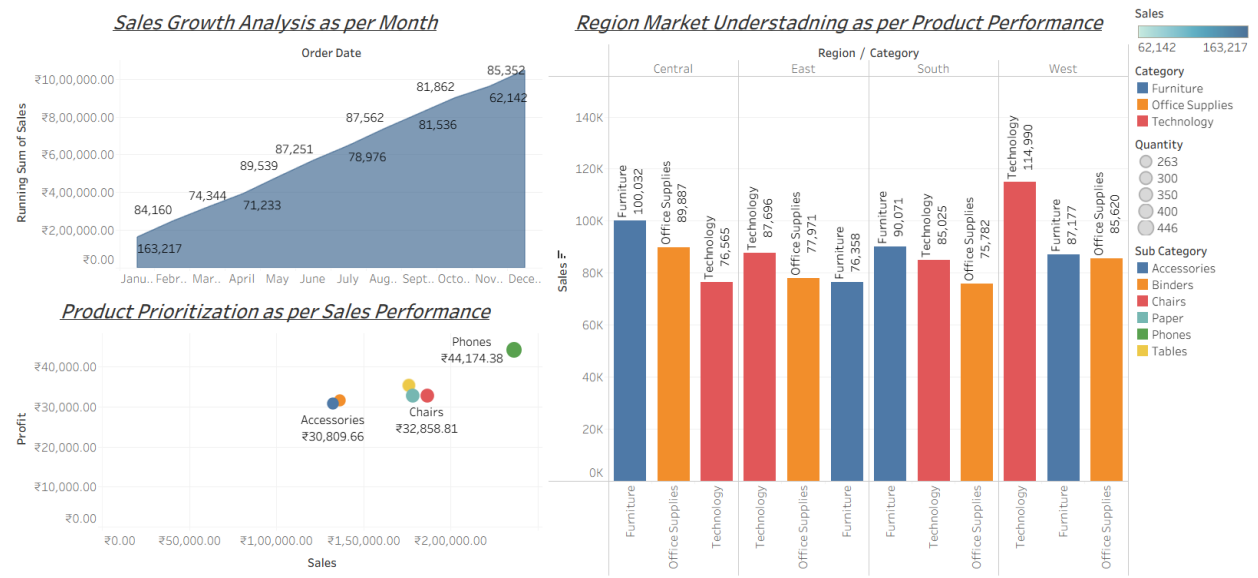
Answer:

Key Insights CEO will see at one glance

Metric	Insight
Top Category	Technology \$364,276
Top Region	West \$287,787 (27.5%)
Growth	Crossed \$1M in 13 months
Best Product	Phones highest Sales + Profit + Quantity
Weak spot	November sales dip every year



Dashboard for Future Growth Prediction



Task 9: Chart Selection Justification Explain why different charts were used for different business questions in this analysis.

Answer:

During this analysis, different chart types were selected based on the nature of the data and the business questions being answered.

- **Bar Chart:** A bar chart was used to compare values across different categories, such as **Sales by Category** and **Profit by Region**. This chart makes it easy to compare performance between groups and quickly identify the highest and lowest values.
- **Line Chart:** A line chart was used to display the **Sales Trend over Time** because it is the most effective way to visualize changes and trends across dates. It helps identify periods of increasing or decreasing sales.

- **Scatter Plot:** A scatter plot was used to analyze the relationship between **Sales and Profit** for each order. This visualization helps identify high-sales orders that generate low profit or even losses, making it useful for profitability analysis.
- **KPI Cards (Text):** KPI cards were used to display important business metrics such as **Total Sales, Total Profit, and Total Orders**. These provide a quick summary of overall business performance without requiring detailed analysis.
- **Interactive Filters:** Filters such as **Region** and **Category** were added to the dashboard to allow users to explore the data dynamically. This enables decision-makers to focus on specific regions or product categories and gain deeper insights.

Task10: Business Insight Summary Summarize two key insights from your dashboard and explain how they can help management make better decisions.

Answer:

Insight 1: Sales and Profit vary across Categories and Regions.

The dashboard shows that some product categories and regions generate higher sales and profits than others. By identifying the best-performing and least-performing areas, management can focus on promoting profitable categories, improve underperforming regions, and allocate resources more effectively.

Insight 2: Sales Trends help identify business performance over time.

The sales trend chart highlights periods of high and low sales throughout the year. This helps management understand seasonal demand, plan inventory more efficiently, and schedule marketing campaigns during slower periods to improve overall sales performance.

Conclusion

These insights enable management to make data-driven decisions by identifying profitable business areas, improving low-performing segments, optimizing resource allocation, and planning future sales and marketing strategies more effectively.